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Cooperative Learning of Shared Latent Dynamics for Distributed Online Optimization

Shivanshu Tripathi STRIP008@UCR.EDU
Department of Electrical and Computer Engineering, UC Riverside, CA, USA
Maziar Raissi MAZIARR@UCR.EDU
Department of Mathematics, UC Riverside, CA, USA

Abstract

We study networked agents that must agree on a decision minimizing the sum of their local costs, where each cost contains unmeasurable stochastic parameters that evolve with linear but unknown dynamics shared by all agents. Each agent receives only finitely many noisy gradient measurements and must then keep tracking the moving minimizer. Our scheme reconstructs the parameters from windows of gradients, learns the shared dynamics cooperatively by averaging instrumental-variable moment matrices over the network, and tracks the minimizer by gradient tracking on forecast gradients. We bound the error of every agent in learning the dynamics and in tracking the minimizer, and show that the learning error, and through it the tracking error, decreases with the number of agents. We demonstrate the approach on two case studies.

Keywords: Online learning, system identification, linear systems finite-sample analysis.

1. Introduction

Online optimization is a fundamental tool for real-time decision making, with applications ranging from the control of robots to context recommendation, in which the cost to be minimized changes over time and its minimizer must be tracked as the cost function changes (e.g., see Simonetto et al., 2020; Dall'Anese et al., 2020; Wensing et al., 2024). While existing methods track a moving minimizer with guarantees when the model of the parameters is known or gradients are available at all times, and learn unknown dynamics with guarantees when the states are measured, they offer no guarantees when the parameters are latent, the data are finite, and the information is dispersed and private, which is the typical situation in networked systems. This poses critical limitations on the deployment of online optimization for multi-agent decision making.

In this paper we propose a cooperative decision making scheme for multi-agent online optimization with the cost function having latent time-varying parameters. These agents cooperatively take a common decision. The latent parameters are observed only through finitely many noisy gradient measurements, so their shared dynamics must be learned in order to keep tracking the optimal decision once the measurements stop. Each agent reconstructs its parameters from windows of measurements, and the agents learn the shared dynamics by averaging instrumental-variable (IV) moment matrices over the network. We then derive bounds on the error of every agent in learning the dynamics and in tracking the minimizer. Our bounds depend on the number of agents, size of the training data, number of consensus rounds, and window length. Finally, we illustrate our findings on two case studies.

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Related work. Online learning in non-stationary environments tracks a moving minimizer of a time-varying cost up to a constant error (Besbes et al., 2015; Mokhtari et al., 2016; Hazan, 2016; Shalev-Shwartz, 2012). Time-varying optimization improves on this with prediction–correction (Simonetto et al., 2016; Simonetto and Dall'Anese, 2017; Fazlyab et al., 2018), contracting dynamics (Davydov et al., 2025), or the internal model principle (Bianchin and Van Scoy, 2026, 2025; Bastianello et al., 2024; Casti et al., 2025), which embeds a known model of the parameter dynamics. Our prior work learned the model for a single agent (Tripathi et al., 2025, 2026). Distributed optimization converges linearly with constant step sizes by gradient tracking (Nedić and Ozdaglar, 2009; Xu et al., 2015; Di Lorenzo and Scutari, 2016; Nedić et al., 2017; Qu and Li, 2018) and its online versions track the moving minimizer (Simonetto et al., 2017; Shahrampour and Jadbabaie, 2018; Carnevale et al., 2023; Notarnicola et al., 2023), but they need the local gradients at every time. In contrast, our agents receive only finitely many gradients. Non-asymptotic system identification (Simchowitz et al., 2018; Oymak and Ozay, 2019; Sarkar and Rakhlin, 2019; Tsiamis and Pappas, 2019; Ziemann and Tu, 2022; Ziemann et al., 2023) and its multi-task, federated, and meta-learning extensions (Wang et al., 2023; Toso et al., 2023; Modi et al., 2024; Toso et al., 2024) show that pooling data from similar systems reduces the sample complexity, but assume measured states, whereas ours are observed through noisy gradients. In contrast to federated learning (McMahan et al., 2017), which aggregates at a server and consensus-based fusion (Xiao et al., 2005), which estimates a static quantity, our agents share only the messages with their neighbors, and the data remains private.

Throughout the paper, $\|\cdot\|$ is the Euclidean norm of a vector and the spectral norm of a matrix, and $\|\cdot\|_F$ is the Frobenius norm. $\sigma_{\min}(\cdot), \lambda_{\min}(\cdot)$, and $\lambda_{\max}(\cdot)$ are the smallest singular value, and the minimum and maximum eigenvalues. $\rho(\cdot)$ is the spectral radius, $(x)_+ := \max\{x, 0\}$, and $\mathrm{col}(v_1, \cdots, v_m)$ stacks the vectors $v_1, \cdots, v_m$.

2. Problem setup

Consider m agents that communicate over an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{1, \ldots, m\}$, and with neighborhoods $\mathcal{N}_i$, and cooperatively solve the time-varying problem:

$$\min_{x \in \mathbb{R}^n} \; F(x, \theta(t)) := \sum_{i=1}^{m} f_i(x, \theta_i(t)), \qquad f_i(x, \theta_i) := f_i^0(x) + g_i(x)^{\mathsf{T}} \theta_i, \tag{1}$$

where $f_i^0 : \mathbb{R}^n \to \mathbb{R}$ and $g_i : \mathbb{R}^n \to \mathbb{R}^p$ are known and twice continuously differentiable, $\theta_i(t) \in \mathbb{R}^p$ is a latent parameter private to agent i, and $\theta(t) := [\theta_1(t)^{\mathsf{T}}, \ldots, \theta_m(t)^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{mp}$. The parameters evolve under unknown linear stochastic dynamics common to all agents as:

$$\theta_i(t+1) = A\theta_i(t) + w_i(t), \qquad w_i(t) \sim \mathcal{N}(0, Q_i), \qquad i \in \mathcal{V}, \tag{2}$$

with unknown $A \in \mathbb{R}^{p \times p}$ and $Q_i \succeq 0$. Agent i has access only to a local gradient oracle:

$$y_i(x, t) = \nabla_x f_i^0(x) + C_i(x)\theta_i(t) + w_{m,i}(t) \tag{3}$$

with $w_{m,i}(t) \sim \mathcal{N}(0, R_i)$, $R_i \succ 0$, and $C_i(x) := \partial g_i^{\mathsf{T}} / \partial x \in \mathbb{R}^{n \times p}$ known; $\theta_i(0)$, $w_i(t)$, and $w_{m,i}(t)$ are mutually independent across agents and time. Since $\nabla_x f_i^0(x)$ is known, agent i can compute $\tilde{y}_i(x, t) := y_i(x, t) - \nabla_x f_i^0(x) = C_i(x)\theta_i(t) + w_{m,i}(t)$.

Data collection: We collect the data for $t \in \{0, \dots, N-1\}$, where agent i queries its oracle at designed exploration points $x_i^{\mathrm{e}}(t)$ and stores $\tilde{y}_i(t) := \tilde{y}_i(x_i^{\mathrm{e}}(t), t)$ and $C_{i,t} := C_i(x_i^{\mathrm{e}}(t))$. During the *tracking phase* $t \geq N$ the oracle is no longer available.

Agent i knows only f_i^0, g_i, R_i and its own data, and exchange messages with its neighbors. The goal is to bound the expected tracking error $\mathbb{E}\|x_i(t) - x^*(t)\|$ of every agent for $t \geq N$, where $x^*(t)$ minimizes (1), and show that it decreases with N and m, when the agents cooperatively estimate the dynamics A of the system. We make the following assumptions:

(A1) There exist $0 < \mu \leq L$ and $\theta_i(t) \in \Theta_i \subseteq \mathbb{R}^p$ for all $t \geq 0$ and $i \in \mathcal{V}$, such that $\nabla_x^2 F(x, \theta) \succeq \mu I_n$ and $\|\nabla_x^2 f_i(x, \theta_i)\| \leq L$ for all $x \in \mathbb{R}^n$, $\theta \in \Theta$, and $i \in \mathcal{V}$.

(A2) A is Schur stable, i.e., spectral radius $\rho(A) < 1$, and invertible, and $(A, Q_i^{1/2})$ is controllable for all $i \in \mathcal{V}$.

(A3) The exploration points lie in a bounded set, and there exist $k_0 \geq \lceil p/n \rceil$ and $\alpha_0 > 0$ with $\sum_{j=0}^{k_0-1} C_{i,t+j}^{\mathsf{T}} C_{i,t+j} \succeq \alpha_0 I_p$ for all $i \in \mathcal{V}$ and $t \in \{0, \dots, N - k_0\}$.

(A4) $\mathcal{G}$ is connected, and $W \in \mathbb{R}^{m \times m}$ is symmetric and doubly stochastic with $W_{ij} > 0$ if and only if $j \in \mathcal{N}_i \cup \{i\}$.

Assumption (A1) makes the cost function $F(\cdot, \theta)$ as μ-strongly convex for every $\theta \in \Theta$, with a unique minimizer $x^*(\theta)$. Since $\nabla_x F(x^*(\theta), \theta') = C(x^*(\theta))(\theta' - \theta)$ with $C(x) := [C_1(x) \cdots C_m(x)]$, for $\theta, \theta' \in \Theta$ we have:

$$\|x^*(\theta') - x^*(\theta)\| \leq L_* \|\theta' - \theta\|, \quad \text{where} \qquad L_* := \frac{1}{\mu} \sup_{\theta \in \Theta} \|C(x^*(\theta))\|. \tag{4}$$

Assumption (A2) guarantees that the dynamics (2) have a unique stationary covariance $\Sigma_{\theta,i} \succ 0$ satisfying $\Sigma_{\theta,i} = A \Sigma_{\theta,i} A^{\mathsf{T}} + Q_i$, and that there exists $C_\rho \geq 1$ and $\rho_A \in (\rho(A), 1)$ with $\|A^j\| \leq C_\rho \rho_A^j$ for all $j \geq 0$. We define $P_i(t) := \mathbb{E}[\theta_i(t) \theta_i(t)^{\mathsf{T}}] \succeq \Sigma_{\theta,i}$. Assumption (A3) is a persistence-of-excitation condition on the exploration sequence with known state-dependent matrix $C_i(x)$. By Assumption (A4), $\sigma := \|W - J\| < 1$, where $J := \mathbf{1}_m \mathbf{1}_m^{\mathsf{T}}/m$.

3. Cooperative identification and tracking

The proposed approach has three steps. First, a local reconstruction step, which computes the Gauss-Markov window estimator of the latent time-varying parameter $\theta_i(t)$. Second, a cooperative identification step, which averages the instrumental-variable moments of all agents over the network. Third, a network optimization step, which is a gradient-tracking iteration driven by the oracle during data collection and by the local forecasts afterwards. The details of each step are presented next. We summarize the overall approach in Algorithm 1.

Step 1 (Local reconstruction of $\theta_i(t)$): For $t < N$, for a window length $k \geq k_0$, agent i stacks the measurements of each window $\{t, \dots, t+k-1\}$, $\bar{y}_{i,t} := [\tilde{y}_i(t)^{\mathsf{T}}, \dots, \tilde{y}_i(t+k-1)^{\mathsf{T}}]^{\mathsf{T}}$, $\bar{C}_{i,t} := [C_{i,t}^{\mathsf{T}}, \dots, C_{i,t+k-1}^{\mathsf{T}}]^{\mathsf{T}}$, $\bar{R}_i := I_k \otimes R_i$, and computes the window estimates from (Tripathi et al., 2026) as

$$\tilde{\theta}_i(t) := K_{i,t} \bar{y}_{i,t}, \qquad K_{i,t} := (\bar{C}_{i,t}^{\mathsf{T}} \bar{R}_i^{-1} \bar{C}_{i,t})^{-1} \bar{C}_{i,t}^{\mathsf{T}} \bar{R}_i^{-1}, \qquad t \in \{0, \dots, N-k\}, \tag{5}$$

where Gauss–Markov estimator neglects the drift of θ_i inside the window. Since $\theta_i(t+j) = \theta_i(t) + (A^j - I)\theta_i(t) + \sum_{l<j} A^{j-1-l} w_i(t+l)$ and $K_{i,t}\bar{C}_{i,t} = I_p$, we have

$$\tilde{\theta}_i(t) = \theta_i(t) + \eta_i(t) + \beta_i(t), \tag{6}$$

with the noise $\eta_i(t) := K_{i,t}\bar{w}_{m,i,t}$, where $\bar{w}_{m,i,t}$ stacks $w_{m,i}(t), \dots, w_{m,i}(t+k-1)$, and the bias $\beta_i(t) := K_{i,t}\bar{b}_{i,t}$, where $\bar{b}_{i,t}$ stacks $C_{i,t+j}(\theta_i(t+j) - \theta_i(t))$, $j = 0, \dots, k-1$. The noise is Gaussian and independent of $\eta_i(s)$ for $|t-s| \geq k$, the bias is zero mean and independent of the noise, and across all agents.

Step 2 (Cooperative identification): Substituting (6) into (2) gives, for every agent:

$$\tilde{\theta}_i(t+1) = A\tilde{\theta}_i(t) + \xi_i(t), \qquad \xi_i(t) := \underbrace{w_i(t)}_{\text{process noise}} + \underbrace{\eta_i(t+1) - A\eta_i(t)}_{\text{noise residual}} + \underbrace{\beta_i(t+1) - A\beta_i(t)}_{\text{bias residual}}. \tag{7}$$

Least squares on (7) is inconsistent, since the regressor and the residual share $\eta_i(t)$ and $\mathbb{E}[\xi_i(t)\tilde{\theta}_i(t)^\mathsf{T}] = -A\Sigma_{\eta,i}(t) \neq 0$. Using the instrument $z_i(t) := \tilde{\theta}_i(t-k)$, local moments are:

$$\hat{M}_{j,i} := \tfrac{1}{N-2k} \textstyle\sum_{t=k}^{N-k-1} \tilde{\theta}_i(t+j) z_i(t)^\mathsf{T}, \qquad j \in \{0, 1\}. \tag{8}$$

Let $M_{j,i} := \mathbb{E}[\hat{M}_{j,i}]$. Since $w_i(t)$ and $\eta_i(t+1) - A\eta_i(t)$ are independent of $z_i(t)$, $M_{1,i} = AM_{0,i} + \Xi_i$ with $\Xi_i := \frac{1}{N-2k}\sum_t \mathbb{E}[(\beta_i(t+1) - A\beta_i(t))z_i(t)^\mathsf{T}]$, for every i with the *same* A; hence the pooled moments $\bar{M}_j := \frac{1}{m}\sum_i M_{j,i}$ satisfy $\bar{M}_1 = A\bar{M}_0 + \bar{\Xi}$, with $\bar{\Xi} := \frac{1}{m}\sum_i \Xi_i$ and $\bar{M}_1\bar{M}_0^{-1}$ identifies A up to the bias residual $\bar{\Xi}\bar{M}_0^{-1}$. The agents compute the pooled sample moments by τ rounds of average consensus and estimate the dynamics as:

$$\hat{M}_j^{(i)}(r+1) = \textstyle\sum_{l \in \mathcal{N}_i \cup \{i\}} W_{il} \hat{M}_j^{(l)}(r), \qquad \hat{A}^{(i)} := \hat{M}_1^{(i)}(\tau)\big(\hat{M}_0^{(i)}(\tau)\big)^{-1}, \tag{9}$$

for $r = 0, \dots, \tau - 1$, with $\hat{M}_j^{(i)}(0) = \hat{M}_{j,i}$.

Step 3 (Network optimization): For $t \geq N$, agent i propagates its last window estimate through the identified dynamics and forms the forecast gradient

$$\hat{\theta}_i(t) := (\hat{A}^{(i)})^{t-(N-k)} \tilde{\theta}_i(N-k), \qquad \hat{y}_i(x,t) := \nabla_x f_i^0(x) + C_i(x)\hat{\theta}_i(t). \tag{10}$$

The predicted minimizer is $\hat{x}^*(t)$. Every agent maintains a decision $x_i(t)$ and a gradient tracker $s_i(t)$ updated by the adapt-then-combine recursion (Xu et al., 2015; Nedić et al., 2017), with sums over $j \in \mathcal{N}_i \cup \{i\}$ as

$$x_i(t+1) = \textstyle\sum_j W_{ij}\big(x_j(t) - \alpha s_j(t)\big), \qquad s_i(t+1) = \sum_j W_{ij}\big(s_j(t) + \gamma_j(t+1) - \gamma_j(t)\big), \tag{11}$$

with step size $\alpha > 0$, $s_i(0) = \gamma_i(0)$, and the gradient signal $\gamma_i(t)$ equal to the oracle gradient $y_i(x_i(t), t)$ for $t < N$ and to the forecast gradient $\hat{y}_i(x_i(t), t)$ for $t \geq N$.

4. Main results

For $t \geq N$ the error of agent i splits as

$$\|x_i(t) - x^*(t)\| \leq \underbrace{\|x_i(t) - \bar{x}(t)\|}_{\text{disagreement}} + \underbrace{\|\bar{x}(t) - \hat{x}^*(t)\|}_{\text{optimization}} + \underbrace{\|\hat{x}^*(t) - x^*(t)\|}_{\text{prediction}}, \tag{12}$$

where $\bar{x}(t) := \frac{1}{m}\sum_i x_i(t)$. By (4)

$$\|\hat{x}^*(t) - x^*(t)\| \leq L_* \sum_i \|\hat{\theta}_i(t) - \theta_i(t)\|,$$

Algorithm 1: Cooperative identification and tracking

Input: f_i^0, g_i, R_i, $\{x_i^{\mathrm{e}}(t)\}_{t<N}$, k, τ, α, $x_i(0)$, $\gamma_i(0) = y_i(x_i(0), 0)$, $s_i(0) = \gamma_i(0)$
for $t = 0, 1, 2, \dots$ **do**
  if $t < N$ **then** query the oracle at $x_i^{\mathrm{e}}(t)$; store $\tilde{y}_i(t)$;
  exchange $x_i(t) - \alpha s_i(t)$ with $\mathcal{N}_i$ and update $x_i(t+1)$ by (11);
  if $t+1 < N$ **then** $\gamma_i(t+1) = y_i(x_i(t+1), t+1)$ `// oracle gradient`;
  if $t+1 = N$ **then**
    compute $\{\tilde{\theta}_i(t')\}_{t' \le N-k}$ by (5) and $\hat{M}_{0,i}, \hat{M}_{1,i}$ by (8);
    run τ consensus rounds and set $\hat{A}^{(i)}$ by (9)
  end
  if $t+1 \ge N$ **then** forecast $\hat{\theta}_i(t+1)$ by (10); $\gamma_i(t+1) = \hat{y}_i(x_i(t+1), t+1)$ `// forecast gradient`;
  exchange $s_i(t) + \gamma_i(t+1) - \gamma_i(t)$ with $\mathcal{N}_i$ and update $s_i(t+1)$ by (11);
end

and, with $H := t - (N-k)$, $\tilde{e}_i := \tilde{\theta}_i(N-k) - \theta_i(N-k)$, and $\theta_i(t) = A^H \theta_i(N-k) + \sum_{j<H} A^j w_i(t-1-j)$, the forecast error of agent i decomposes as

$$\hat{\theta}_i(t) - \theta_i(t) = \underbrace{\big((\hat{A}^{(i)})^H - A^H\big)\tilde{\theta}_i(N-k)}_{\text{identification}} + \underbrace{A^H \tilde{e}_i}_{\text{reconstruction}} - \underbrace{\textstyle\sum_{j<H} A^j w_i(t-1-j)}_{\text{process noise}}. \tag{13}$$

We bound each error in the subsequent subsections. All proofs of lemma's and theorem's are given in the supplementary material.

4.1. Reconstruction and cooperative identification

The estimate $\tilde{\theta}_i(t)$ in (5) has a reconstruction error $\mathbb{E}\|\tilde{\theta}_i(t) - \theta_i(t)\|_2$.

Lemma 1 (Reconstruction error) *Let (A2)–(A3) hold, $\Gamma_i := \sup_{x \in \mathbb{R}^n} \|C_i(x)\|_2$, $B_{\theta,i} := \sqrt{\mathrm{tr}(\Sigma_{\theta,i})}$, and $k \ge k_0$. For all $i \in \mathcal{V}$ and $t \in \{0, \dots, N-k\}$:*

1. *$\bar{C}_{i,t}^{\mathsf{T}} \bar{R}_i^{-1} \bar{C}_{i,t} \succeq \alpha_{k,i} I_p$ with $\alpha_{k,i} := \lfloor k/k_0 \rfloor \alpha_0 / \lambda_{\max}(R_i)$.*
2. *The covariance $\Sigma_{\eta,i}(t)$ of $\eta_i(t)$ satisfies $\Sigma_{\eta,i}(t) \preceq \alpha_{k,i}^{-1} I_p$.*
3. *For $\chi_i^2 := 2k_0 \lambda_{\max}(R_i)/(\alpha_0 \lambda_{\min}(R_i))$, the reconstruction error satisfies*

$$\big(\mathbb{E}\|\tilde{\theta}_i(t) - \theta_i(t)\|^2\big)^{1/2} \le \sqrt{p/\alpha_{k,i}} + \Gamma_i C_\rho \chi_i \Big(\tfrac{k}{\sqrt{3}} \|A - I\| B_{\theta,i} + \sqrt{\tfrac{k}{2}\,\mathrm{tr}(Q_i)}\Big) := \epsilon_i^{\mathrm{w}}. \tag{14}$$

We identify the common dynamics A using the sample moments (8), which every agent computes locally and then averages. The moment of agent i is given as $M_{0,i} = A^k \Sigma_{\theta,i} + \Delta_{0,i}$, where $A^k \Sigma_{\theta,i} = \mathbb{E}[\theta_i(t)\theta_i(t-k)^{\mathsf{T}}]$ is the correlation between the regressor $\tilde{\theta}_i(t)$ and the lagged instrument $\tilde{\theta}_i(t-k)$, and $\Delta_{0,i}$ collects the terms involving the windowing bias β_i. The pooled moment is $\bar{M}_0 = A^k \bar{\Sigma}_\theta + \bar{\Delta}_0$ with $\bar{\Sigma}_\theta := \frac{1}{m}\sum_i \Sigma_{\theta,i}$ and $\bar{\Delta}_0 = \frac{1}{m}\sum_i \Delta_{0,i}$. We define the invertability margin $\nu_k := \sigma_{\min}(A^k \bar{\Sigma}_\theta) > 0$ of $\bar{M}_0$, and the heterogeneity $D_M := \max_{j \in \{0,1\}, l \in \mathcal{V}} \|M_{j,l} - \bar{M}_j\|_2$, which vanishes for identical agents.

The estimate $\hat{A}^{(i)}$ differs from A for three reasons. First, the moments are computed from finitely many samples (finite sample error). Second, after τ consensus rounds, agent i knows the network average only up to a consensus error of order σ^τ. Third, the error due to bias residual $\bar{\Xi}\bar{M}_0^{-1}$ contributes an error that does not vanish with N.

Theorem 2 (Identification error) *Let Assumptions (A2)–(A4) hold, $k \geq k_0$, $\delta \in (0, 1/2]$, $N \geq N_0(\delta/(4m))$, $\|\bar{\Delta}_0\| \leq \nu_k/2$, and $\epsilon^{\mathrm{m}}(N, \tau, \delta) \leq \nu_k/8$, where*

$$\epsilon^{\mathrm{m}}(N, \tau, \delta) := c\sqrt{\log(4m/\delta)/N}\left(m^{-1/2} + \sigma^{\tau}\right) + \sqrt{m}\,\sigma^{\tau} D_M. \tag{15}$$

where $c > 0$ is the concentration constant *of the sample moments, independent of N, m and δ such that, for every $j \in \{0, 1\}$, every weight vector $\omega \in \mathbb{R}^m$ and every $\delta \in (0, 1)$, we have*

$$\left\| \textstyle\sum_{l=1}^{m} \omega_l \big(\hat{M}_{j,l} - M_{j,l}\big) \right\| \leq c\,\|\omega\| \sqrt{\tfrac{\log(1/\delta)}{N}}. \tag{16}$$

Then there is an event $\mathcal{E}_\delta$ with $\mathbb{P}(\mathcal{E}_\delta) \geq 1 - \delta$ on which, for all $i \in \mathcal{V}$ simultaneously,

$$\|\hat{A}^{(i)} - A\| \leq \underbrace{C_A(1 + \sqrt{m}\sigma^{\tau})\sqrt{\frac{\log(4m/\delta)}{mN}}}_{\textit{finite-sample error}} + \underbrace{C_W\sqrt{m}\sigma^{\tau} D_M}_{\textit{heterogeneity}} + \underbrace{C_B(k\|A - I\|_2 + \sqrt{k\,\mathrm{tr}(\bar{Q})})}_{\textit{bias residual}}$$
$$=: \epsilon^{id}(\delta),$$

where $\bar{Q} := \frac{1}{m}\sum_i Q_i$, and the constants C_A, C_B, C_W are independent of N, τ and δ.

4.2. Tracking

For $t \geq N$ the oracle is no longer available and agent i replaces its gradient by the forecast gradient $\hat{y}_i(x, t)$ of (10), which is the gradient of the *forecast cost* $\hat{f}_i(x, t) := f_i^0(x) + g_i(x)^{\mathsf{T}}\hat{\theta}_i(t)$. The algorithm tracks the minimizer $\hat{x}^*(t)$ of $\sum_i \hat{f}_i(\cdot, t)$. Let $x(t) := \mathrm{col}(x_1(t), \ldots, x_m(t))$, $s(t) := \mathrm{col}(s_1(t), \ldots, s_m(t))$, $\hat{\theta}(t) := \mathrm{col}(\hat{\theta}_1(t), \ldots, \hat{\theta}_m(t))$, and $\bar{s}(t) := \frac{1}{m}\sum_i s_i(t)$. We define:

$$\begin{aligned} a_t &:= \|x(t) - \mathbf{1}_m \otimes \bar{x}(t)\|_2 \quad \text{(iterate disagreement)}, \\ b_t &:= \sqrt{m}\,\|\bar{x}(t) - \hat{x}^*(t)\|_2 \quad \text{(optimization error)}, \\ c_t &:= \|s(t) - \mathbf{1}_m \otimes \bar{s}(t)\|_2 \quad \text{(tracker disagreement)}, \\ d_t &:= \|\hat{\theta}(t+1) - \hat{\theta}(t)\|_2 \quad \text{(forecast drift)}, \end{aligned} \tag{17}$$

with $e_t := [a_t \;\; b_t \;\; c_t]^{\mathsf{T}}$. The first two are the disagreement and optimization terms of (12), c_t measures how well the trackers follow the average gradient, and d_t bounds the motion of the predicted minimizer, since $\|\hat{x}^*(t+1) - \hat{x}^*(t)\| \leq L_* d_t$ by (4).

Lemma 3 (Gradient tracking) *Let (A1)–(A4) hold, let $\kappa := mL/\mu$, $\Gamma := \max_{i \in \mathcal{V}} \Gamma_i$, and let $\alpha \in (0, \alpha_{\max}]$, where $\alpha_{\max} := \min\left\{\frac{(1-\sigma)^2}{4L}, \; \frac{1}{(\kappa+1)L}\right\}$. Then there exist $\rho_{\mathrm{gt}} \in (0, 1)$ and $c_{\mathrm{gt}} > 0$, such that for all $i \in \mathcal{V}$ and $t \geq N$,*

$$\|x_i(t) - \hat{x}^*(t)\|_2 \leq c_{\mathrm{gt}}\Big(\rho_{\mathrm{gt}}^{t-N}\|e_N\|_2 + \sum_{q=N}^{t-1} \rho_{\mathrm{gt}}^{t-1-q} d_q\Big). \tag{18}$$

The iterates forget their initial state at the rate ρ_{gt} and lag behind the predicted minimizer by a weighted sum of past drifts. The drift is produced by forecasts as next theorem shows.

The tracking error of agent i thus has three sources. First, the network needs time to reach consensus and to optimize. Second, the predicted minimizer moves while the forecasts are propagated, so the iterates lag behind it. Third, the forecasts are wrong, by the identification, reconstruction, and process-noise terms of (13), which shifts the minimizer away from $x^*(t)$. Since the identification results hold on an event $\mathcal{E}_\delta$. For a random variable Z, we have $\mathbb{E}_\delta[Z] := \mathbb{E}[Z\mathbb{I}_{\mathcal{E}_\delta}]$ for $Z \geq 0$, so that $\mathbb{E}[Z \mid \mathcal{E}_\delta] \leq \mathbb{E}_\delta[Z]/(1 - \delta)$.

Theorem 4 (Distributed tracking error) *Let Assumptions (A1)–(A4) and the hypotheses of Theorem 2 and Lemma 3 hold, and let* $\hat{\rho} := \rho_A + C_\rho \epsilon^{\mathrm{id}}(\delta) < 1$ *and* $\rho_\star := \max\{\rho_{\mathrm{gt}}, \hat{\rho}\}$. *Then, for every agent* $i \in \mathcal{V}$ *and every* $t \geq N$,

$$\mathbb{E}_\delta \|x_i(t) - x^*(t)\| \leq \underbrace{c_{\mathrm{gt}} \rho_{\mathrm{gt}}^{t-N} \mathbb{E}_\delta \|e_N\|}_{\text{(i) optimization error}} + \underbrace{c_{\mathrm{gt}} C_\rho \hat{\rho}^k (t-N) \rho_\star^{t-N-1} U_N}_{\text{(ii) forecast drift}} + \underbrace{L_* \textstyle\sum_{j=1}^m \epsilon_j^{\mathrm{f}}(N, t; \delta)}_{\text{(iii) forecast error}}, \tag{19}$$

where $U_N := \big(\|A - I\| + \epsilon^{\mathrm{id}}(\delta)\big) \sum_{j=1}^m \big(B_{\theta,j} + \epsilon_j^{\mathrm{w}}\big)$, $\mathbb{E}_\delta \|e_N\| < \infty$, *and* $\epsilon_j^{\mathrm{f}}(N, t; \delta)$ *bounds the forecast error of agent* j *with* $H = t - (N - k)$ *and* $\sigma_{H,j}^2 := \operatorname{tr} \sum_{l<H} A^l Q_j (A^l)^\mathsf{T} \leq \operatorname{tr} \Sigma_{\theta,j}$,

$$\underbrace{\mathbb{E}_\delta \|\hat{\theta}_j(t) - \theta_j(t)\|}_{\text{forecast error}} \leq \underbrace{C_\rho^2 H \hat{\rho}^{H-1} \big(B_{\theta,j} + \epsilon_j^{\mathrm{w}}\big) \epsilon^{\mathrm{id}}(\delta)}_{\text{identification}} + \underbrace{C_\rho \rho_A^H \epsilon_j^{\mathrm{w}}}_{\text{reconstruction}} + \underbrace{\sigma_{H,j}}_{\text{process noise}}. \tag{20}$$

5. Numerical experiments

We evaluate the proposed methodology on two case studies: (i) charging-cart placement for patrolling UAVs, simulated in Webots, and (ii) decentralized recommendation with regional servers. The project page and source code are publicly available at video and code.

5.1. A charging cart for patrolling UAVs

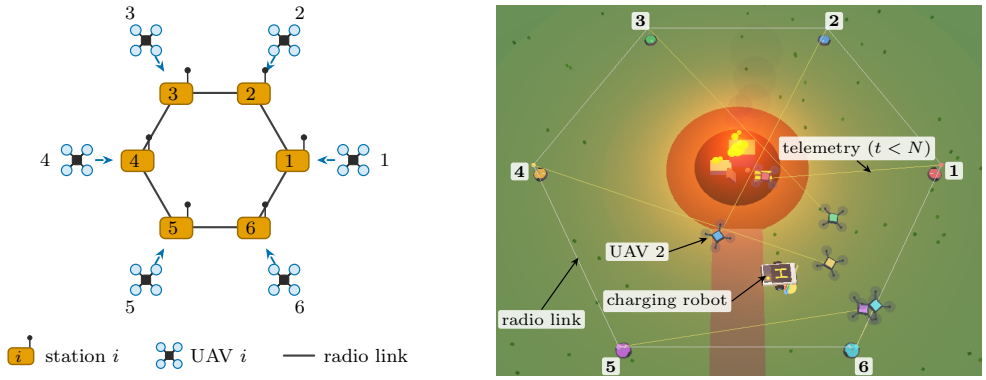
Figure 1 illustrates the setting and the Webots simulation setup. During the response to a wildfire, six agencies (police, fire department, emergency medical services, search and rescue, utilities, and civil protection) each fly one patrol UAV over the site. UAV i follows a fixed patrol route $p_i(t)$, known only to its agency, whose ground station receives the position and telemetry of that UAV only. The agencies share neither telemetry, nor video, nor routes. The cellular network is limited, where each station exchanges messages only with its nearest stations. A ground robot carries a charging station for the UAVs; a UAV that needs power flies to the cart, charges, and returns to its route. The stations must agree on where the cart should be, and move it as the UAVs move. The cart is simulated as a wheeled robot. The position of UAV i is its planned position plus a deviation,

$$b_i(t) = p_i(t) + \theta_i(t), \qquad \theta_i(t+1) = A\,\theta_i(t) + w_i(t).$$

The deviation $\theta_i(t)$ is unknown and wind gusts $w_i(t)$ push the UAV off its route. The deviations of all UAVs share the dynamics $A = \begin{bmatrix} 0.979 & -0.148 \\ 0.148 & 0.979 \end{bmatrix}$, which is unknown to the stations. Putting the cart at x costs UAV i the squared distance to it, weighted by a known constant h_i, denoting the rate at which that UAV needs power. The cost function is given as

$$f_i(x, \theta_i) = h_i \|x - b_i(t)\|^2 = h_i \|x - p_i(t) - \theta_i\|^2.$$

The team cost is smallest at the weighted centroid $x^\star(t) = \sum_i h_i b_i(t) / \sum_i h_i$, which moves with the UAVs. We consider $f_i^0(x) = h_i \|x - p_i(t)\|^2$ and $g_i(x) = -2h_i(x - p_i(t))$, so $C_i = -2h_i I_2$ is invertible and Assumption (A3) holds with $k_0 = 1$. A noisy gradient of f_i at any x is a noisy position of UAV i, which station i receives once per second during data collection ($t < N$). At time $t = N$, data collection stops and no station receives the position of its UAV, and each must forecast it to keep the cart in place. The UAVs start on average 50 m off their routes, gusts displace them by 0.5 m per second, and the received positions



(a) Ground stations and radio links (b) Webots, during data collection

Figure 1: The setting of Section 5.1. (a) Station i receives the telemetry of UAV i only, while $t < N$, and talks only to its two neighbors on the ring. (b) The Webots simulation setup with each UAV having the color of its station.

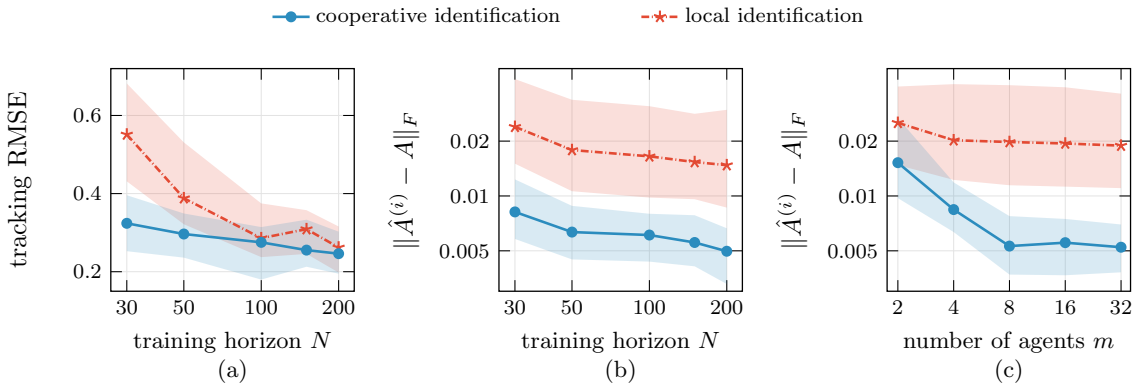
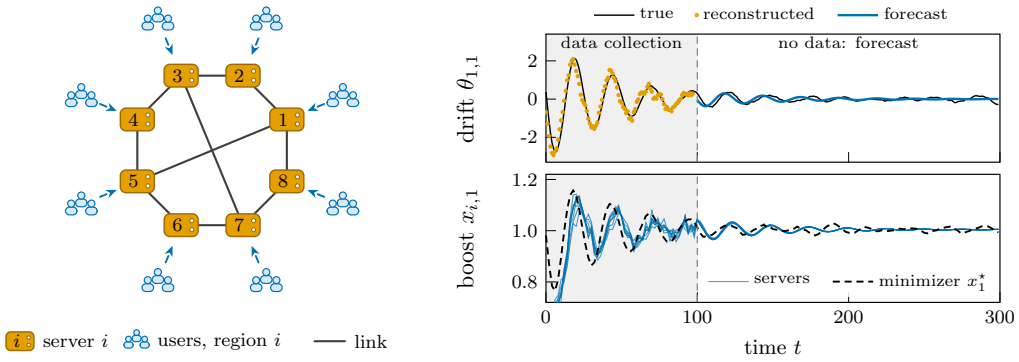


Figure 2: Cooperative identification (solid blue) and local identification (dash-dotted red) for the setting of Section 5.1, over 50 Monte Carlo trials with a tracking horizon of 200 steps. (a) Tracking RMSE vs the training horizon N, (b) identification error $\|\hat{A}^{(i)} - A\|_F$ vs the training horizon N, and (c) identification error versus the number m of identical agents ($h_i = 1$) and $N = 100$.

carry errors of 5 to 15 m. Each station talks only to the nearest neighbor in the network of Figure 1(a).

The iterates start at $x_i(0) = 0$ with $s_i(0) = \gamma_i(0)$. Cooperative identification has $\tau = 3$ consensus rounds and $\tau = 0$ for local identification. Fig. 2(a) shows the tracking RMSE $\left(\frac{1}{150m}\sum_{t=N+50}^{N+199}\sum_{i=1}^{m}\|x_i(t) - x^*(t)\|^2\right)^{1/2}$ for different values of N. We observe that the tracking error decreases with N. Fig. 2(b) shows that the cooperative system identification is 2.7–3 times more accurate than the local identification by single station for every N. The cooperative error decreases from 0.008 to 0.005 as N grows. Fig. 2(c) shows that the error for cooperative identification decreases as agents increase (as $1/\sqrt{m}$ from Theorem 2) but saturates for 16 or more agents, where σ reaches 0.85 and 0.96 and σ^τ is no longer small. However, for local identification, the error remains the same.



(a) Regional servers and backbone links

(b) One trial, $N = 100$

Figure 3: The setting of Section 5.2. (a) Network configuration with server i receiving only the engagement data of region i for $t < N$, and exchanges messages with its backbone neighbors. (b) Top figure shows the drift $\theta_{1,1}$ for region 1 and genre 1, its reconstruction by server 1 ($t < N$), and its forecast with the cooperative estimate of A ($t \geq N$). Bottom figure shows the boosts $x_{i,1}$ of genre 1 at the eight servers and the minimizer $x_1^\star$ of the team cost.

5.2. Decentralized recommendation with regional servers

Figure 3 illustrates the setting with servers, users and links. A content platform serves its users from $m = 8$ regional servers. Each regional server collects the data of the region, and there is no central aggregator, as in decentralized federated learning (Koloskova et al., 2020). Each server exchanges messages only with two or three neighbors. Only the ranking model is shared, so the servers must agree on common top 3 boosts $x \in \mathbb{R}^3$. The interest of the users of region i in genre ℓ is a known average plus a drift,

$$u_{i,\ell}(t) = \bar{u}_{i,\ell} + \theta_{i,\ell}(t), \qquad \theta_i(t+1) = A\,\theta_i(t) + w_i(t).$$

The drift $\theta_i(t)$ is unknown and private to server i, and the noise $w_i(t)$ models short-term fluctuations of taste. The interests of all regions follow the same trend dynamics

$$A = \begin{bmatrix} 0.9495 & -0.2425 & 0 \\ 0.2425 & 0.9495 & 0 \\ 0 & 0 & 0.95 \end{bmatrix},$$

which is unknown to the servers. The dynamics shows a damped oscillation shared by the first two genres and a fading interest in the third. Under a boost x_ℓ, a user engages with an item of genre ℓ with probability $P(x_\ell)$, where $P(x_\ell) = 1/(1+e^{-z})$ is the logistic function, so the negative likelihood of the engagement is $(-\log P(x_\ell)) = softplus(-x_\ell)$. The server i weights the loss by interest of the user and ridge penalty with weight $\lambda_i > 0$, which keeps the boosts small:

$$f_i(x, \theta_i) = \frac{\lambda_i}{2}\|x\|^2 + \sum_{\ell=1}^{3} u_{i,\ell}(t)\,\mathrm{softplus}(-x_\ell),$$

We consider $f_i^0(x) = \frac{\lambda_i}{2}\|x\|^2 + \sum_\ell \bar{u}_{i,\ell}\,\mathrm{softplus}(-x_\ell)$ and $g_i(x) = \mathrm{softplus}(-x)$ (elementwise), so $C_i(x) = -\operatorname{diag}(P(-x))$ is invertible for every x and Assumption (A3) holds with $k_0 = 1$. The noise $w_i(t)$ has standard deviation 0.02, and the gradients carry noise of standard deviation 0.15. The network configuration is shown in Fig. 3(a).

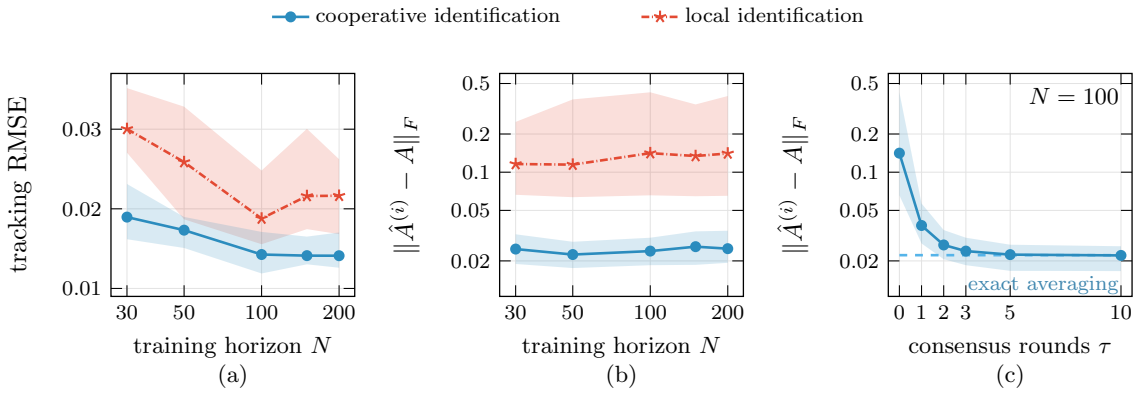


Figure 4: Cooperative identification with $\tau = 3$ (solid blue) and local identification $\tau = 0$ (dash-dotted red) for the setting of Section 5.2, over 50 Monte Carlo trials with a tracking horizon of 200 steps. (a) Tracking RMSE vs the training horizon N, (b) identification error $\|\hat{A}^{(i)} - A\|_F$ versus N and (c) identification error versus the number of consensus rounds.

We run 50 Monte Carlo trials with the same tracking horizon, initialization, simulation setting, and tracking RMSE as in (A). Figure 3(b) shows one trial with $N = 100$. Fig. 4(a) shows the tracking RMSE with cooperative identification is 24–37% lower than with local identification for every N. Fig. 4(b) shows the error in identification of A. Cooperation reduces it 4.7–5.9 times compared to the local estimates. The cooperative error stays between 0.022 and 0.026 for every N. Fig. 4(c) varies the number of consensus rounds at $N = 100$ ($\sigma = 0.68$). The identification error decreases from 0.141 for $\tau = 0$ to 0.022 for $\tau = 5$.

Summary. In both case studies, we observe that the cooperation reduces the identification error by the factor $\sqrt{m}$ of Theorem 2, or more when local identification is unreliable. The cooperative error settles at the bias floor set by k and $\|A - I\|$ as shown in Theorem 4.

6. Conclusion

We studied online optimization over a network of agents whose local costs depend on private latent parameters with shared unknown stochastic dynamics, observed only through finitely many noisy gradients. Averaging instrumental-variable moments over the network learns the dynamics and estimates the minimizer. The simulations confirm the $\sqrt{m}$ gain in cooperative estimation, and show that a few consensus rounds recover the dynamics, and show that cooperation matters most when data are scarce. Future work includes considering directed or time-varying graphs, and considering non-linear dynamics.

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